Weight-aware imputation for policy microsimulation

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Abstract

Policy microsimulation models rely on imputation to attach variables one survey observes to records from another, and the imputed file is then reweighted downstream: calibrated to administrative totals, filtered to subpopulations, stressed under reform scenarios. Fitting imputation models to survey records while ignoring their design weights biases imputed conditionals toward the sample rather than the population. We present the imputation operator of MICROCOSM, POLICYENGINE's open-source microdata stack: a regime-gated, sequentially chained, weighted-bootstrap quantile-regression-forest estimator whose interface makes an unweighted fit impossible to request by accident. To evaluate it we formalize a *population-view harness*: one latent population observed through survey views, with any candidate weighted file scored by projecting it through each view against that survey's holdout. Benchmarked against standard survey-imputation methods on four task families, weighting and chaining each earn their place where their failure modes live. Fit without weights, the identical estimator is six times worse on net-worth marginal fit and imputes a 99th percentile 1.9 to 2.3 times the holdout's — an inflation that capped joint-geometry metrics miss entirely and only an uncapped tail axis detects. The gated estimator holds dividend zero-share error to 0.6 percentage points, against 4.4 for a standard ungated forest imputer. Chaining is worth 5.8 points of classifier two-sample AUC once four imputed components must satisfy a balance-sheet identity. The estimator is available standalone as `microcosm-fit`, and every number in the paper regenerates from the accompanying repository.

Keywords: microsimulation, imputation, survey weights, quantile regression forests, statistical matching

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1 Introduction

Microsimulation models of tax and transfer policy require individual records that span domains no single survey covers (Bourguignon and Spadaro, 2006; Sutherland and Figari, 2013). The Current Population Survey observes employment and income but not wealth; the Survey of Consumer Finances observes wealth in detail but lacks the CPS's coverage and program detail; administrative tax records observe income components precisely but only for filers, and with narrow demographics. Model builders bridge these gaps with imputation: fit a conditional model of the missing variables on a donor survey, then draw values for the receiver's records (Little and Rubin, 2002; D'Orazio et al., 2006).

Two properties of this setting distinguish it from generic missing-data problems. First, both files are *weighted samples*: each record stands in for a different number of population units, and the quantity of interest is a population distribution, not a sample one. An imputation model fit to unweighted records recovers the sample conditional, which differs from the population conditional exactly when the design (or a prior calibration) correlates weights with outcomes. Second, the imputed file is not an endpoint. It is *reweighted downstream*: calibrated to administrative margins (Deville and Särndal, 1992; Hainmueller, 2012), filtered to subpopulations, stressed under reforms. An imputation that looks acceptable under the weights it shipped with can therefore fail under weights it never saw. A rare donor record carrying an extreme value at near-zero weight is invisible in the shipped aggregates; a later reweighting that assigns it mass turns it into a distortion of the population total. We refer to such records as landmines, and we measure exposure to them directly with a reweighting stress test (Section 5.2, Table 4).

This paper presents and evaluates the imputation operator of MICROCOSM,[1] POLICYENGINE's open-source microdata stack (PolicyEngine, 2026). The estimator combines three design choices, each aimed at one of the failure modes above. A *weighted bootstrap* materializes survey weights into the training data of each forest, so the learned conditionals are population conditionals. *Regime gates* classify each record's sign class (negative, zero, positive) before magnitudes are modeled, so zero-inflated and sign-mixed variables — ubiquitous in economic microdata (Mullahy, 1986; Lambert, 1992) — keep their mass structure. *Sequential chaining* conditions each imputed variable on the ones already drawn, so joint structure among targets survives the imputation (Raghunathan et al., 2001;

[1] MICROCOSM was first released under the name Populace. Artifact and release identifiers created before the rename keep the legacy `populace` prefix, and the reproducibility pins in the data-availability section reference the repository at a pre-rename commit.

Van Buuren and Groothuis-Oudshoorn, 2011). The interface enforces the first choice: fits are weighted by construction, and an unweighted fit must be requested explicitly.

We make three contributions:

1. **A weight-aware imputation estimator, available standalone.** We describe the estimator and its interface contract, and release it as `microcosm-fit`, installable and usable on plain data frames independently of the rest of the MICROCOSM stack.

2. **The population-view harness.** We frame the evaluation question as one latent population observed through survey *views* — variable subsets, designs, measurement idioms — and score any candidate weighted file by projecting it through each view against that survey's holdout. The score has four blocks: a strictly proper joint score (Gneiting and Raftery, 2007; Székely and Rizzo, 2013), a support-based coverage axis that is invariant to reweighting of the candidate by definition (Naeem et al., 2020), a weighted classifier two-sample test, and an uncapped tail block on imputed columns. We added the tail block after finding that capped sample-geometry metrics are blind to twofold errors in the extreme tail, where economic variables carry their policy weight. The harness works for any generator, is non-self-referential (holdouts appear nowhere upstream), and cleanly separates what record generation must get right from what calibration can later change.

3. **A benchmark against standard practice with attribution, plus a reweighting stress test.** We compare the estimator against unweighted and weighted quantile regression forests (Meinshausen, 2006), ordinary least squares imputation (von Hippel, 2013), linear quantile regression (Koenker and Bassett, 1978), and nearest-neighbour hot-deck statistical matching (Andridge and Little, 2010; D'Orazio et al., 2006), under a paired, repeated holdout protocol. Ablations knock out weighting and chaining one at a time, so each gets its own attribution; gating is compared against an ungated forest baseline. A closed-form fragility diagnostic measures each method's worst-case single-record exposure under bounded reweightings of the imputed file — the landmine failure mode.

The policy stakes are concrete. Supplemental Security Income conditions eligibility on countable resources (Social Security Administration, 2024), so simulating its baseline caseload on the CPS — or reforms such as the SSI Savings Penalty Elimination Act, which would raise the resource limits (119th United States Congress, 2025) — is impossible without imputed wealth. The composition of the simulated caseload is exactly a functional

of the imputed demographics–wealth joint. This paper does not simulate the SSI caseload; it measures the property any such simulation depends on. The population-view experiment of Section 4.2 scores the imputed joint directly: it asks whether households assembled from CPS demographics and imputed wealth are distributionally exchangeable with households the SCF actually observed.

The rest of the paper proceeds as follows. Section 2 situates the estimator in the survey-imputation literature. Section 3 specifies the estimator, the ablation design, the metrics, and the protocol. Section 4 describes the four benchmark tasks and their data. Section 5 reports results, Section 6 discusses limitations, and Section 7 concludes.

2 Background

2.1 Imputation methods in survey practice

Survey agencies and microsimulation teams draw on a small set of imputation families. *Hot-deck and statistical matching* methods donate observed values from a matched donor record; they preserve marginal distributions by construction because every imputed value is a real donor value (Andridge and Little, 2010; D'Orazio et al., 2006), and remain the standard data-fusion tool in European microsimulation practice (Sutherland and Figari, 2013). *Regression-based* methods impute from a fitted conditional model — ordinary least squares with residual draws (von Hippel, 2013), linear quantile regression (Koenker and Bassett, 1978; Koenker, 2005) — and extend to multiple imputation for uncertainty propagation (Rubin, 1987; Kennickell, 1998). *Tree-ensemble* methods, in particular quantile regression forests (Meinshausen, 2006; Breiman, 2001), estimate full conditional distributions non-parametrically. They are the default in POLICYENGINE's data pipelines (Ghenis, 2018; Woodruff and Ghenis, 2024) because they capture nonlinear structure without per-variable specification. The `microimpute` package (PolicyEngine, 2025) implements these families under one interface and selects among them by cross-validation; its central finding — no family dominates across datasets — motivates empirical comparison on each new task rather than a prescribed method.

2.2 Chained imputation and joint structure

Imputing several variables one at a time, each conditional only on shared predictors, destroys the dependence among imputed variables; sequential (chained-equations) imputation conditions each variable on those already imputed, retaining joint structure (Raghunathan et al.,

2001; Van Buuren and Groothuis-Oudshoorn, 2011). In data fusion the same issue appears as the conditional independence assumption of statistical matching: matched files preserve marginals but attenuate cross-source correlations (D'Orazio et al., 2006). Synthetic-data generation faces the identical trade-off at whole-file scale (Rubin, 1993).

2.3 Weights in imputation

Design weights enter imputation practice unevenly. The model-based literature often treats weights as a nuisance — if the model is correctly specified and the design variables are conditioned on, weighting is unnecessary — while the design-based tradition weights every estimator (Little and Rubin, 2002). In production microdata pipelines the model is never correctly specified and the design variables are never fully available, so the choice is consequential: an unweighted fit targets the sample conditional, and any correlation between weights and outcomes within predictor cells propagates into the imputed population distribution. Tree ensembles add a mechanical subtlety: a fully-grown forest's *predictive distribution* is a leaf-membership distribution, which per-record `sample_weight` arguments do not reweight. Honoring weights therefore requires materializing them into the training data — the weighted bootstrap of Section 3. Zero-inflated targets add a second subtlety: weighting the zero/nonzero gate by resampling can delete a rare sign class outright, so the gate must be weighted directly. We are not aware of prior survey-imputation work that combines weight-aware forest training, sign-regime gating, and chaining in one estimator; documenting and testing that combination is the gap this paper fills.

2.4 Evaluating imputed files

Because an imputation is a draw from an estimated conditional distribution, pointwise accuracy metrics understate quality; the relevant question is distributional. We report pinball loss, from the quantile-loss tradition (Koenker, 2005; Meinshausen, 2006), and Wasserstein distance; as computed here both score marginal fit (Section 3.4). For the joint, the evaluation perspective this paper formalizes — one latent population, surveys as partial views of it, candidates scored through each view against held-out survey samples — has ancestry in several literatures that have not, to our knowledge, been assembled into an operational benchmark: the survey-statistics treatment of multiple sources observing one population (Lohr and Raghunathan, 2017), the missing-data view of fusion as block missingness on a single file (Little and Rubin, 2002; Rässler, 2002), strictly proper scoring of distributional claims against realized samples (Gneiting and Raftery, 2007; Székely and Rizzo, 2013), classifier

two-sample tests (Lopez-Paz and Oquab, 2017), and the precision–recall–density–coverage geometry of generative-model evaluation (Naeem et al., 2020). Population-synthesis practice validates against input marginals rather than held-out views. Section 3.3 operationalizes the combination; the reweighting fragility diagnostic of Section 3.4 is, to our knowledge, new.

3 Methodology

3.1 The estimator

`microcosm-fit` models $P(\text{targets} \mid \text{predictors})$ with a quantile-regression-forest estimator (Meinshausen, 2006; Johnson, 2024) wrapped in three mechanisms.

Weighted bootstrap. Random forests cannot honor a per-record `sample_weight` in their predictive distribution: a fully grown leaf holds individual training rows, and weighting the impurity criterion does not change which values a quantile query reads out. `microcosm-fit` therefore materializes weights into the data: before each forest is grown, training rows are resampled with replacement with probability proportional to weight, so leaf distributions reflect the weighted population. Draws from the fitted forest then sample the *weighted* conditional.

Regime gates. Economic variables are rarely smooth. Most adults hold zero dividend income; net worth is negative for some households, exactly zero for others, and spans several orders of magnitude when positive. A single regressor fit across such a target either drops a tail or interpolates across the empty gap at zero; hurdle-style two-part models are the classical remedy (Mullahy, 1986; Lambert, 1992). `microcosm-fit` generalizes the two-part idea to the target's *sign support* — whichever of $\{\text{negative}, \text{zero}, \text{positive}\}$ actually occur — and imputes in four steps:

1. *Detect the regime.* Read which sign classes occur in the training data. Detection is deliberately unweighted: which signs *exist* is a fact about the variable, not about the sampling design, and a rare but real class (negative net worth, say) must not vanish because its weight share is small.

2. *Fit the gate.* A classifier learns the probability of each sign class given the predictors. The gate is weighted directly through its `sample_weight` argument rather than by resampling the training rows — resampling could delete a rare sign class outright, exactly the failure step 1 exists to prevent.

3. *Fit magnitudes within each sign.* A separate quantile-regression forest models the magnitude of each nonzero sign class, so no forest ever straddles the gap at zero.

4. *Draw.* For each receiver row, sample a sign class from the gate's predicted probabilities — not the modal class, which would collapse the zero mass — and, if the class is nonzero, draw a magnitude from that class's forest.

For dividend income the regime is $\{0, +\}$ and the gate is a zero hurdle; for net worth it is $\{-, 0, +\}$ and the gate is three-way. The gate's probabilities are those of the classifier as fitted on the training rows: the only behavior at the version evaluated here, and still the package's default. For a rare sign class such a gate can be overconfident on rows it has not seen, and later versions of `microcosm-fit` add an opt-in out-of-fold calibration of the gate for that case (PolicyEngine, 2026). The tasks in this paper contain no rare sign class on which the methods differ, so the gate's behavior on rare classes is not evaluated here. Section 5.2 compares the gated estimator with ungated baselines on zero-inflated targets.

Sequential chaining. Targets are imputed in order, each conditioning on the predictors plus the targets already drawn (Raghunathan et al., 2001; Van Buuren and Groothuis-Oudshoorn, 2011), so dependence among imputed variables survives.

Interface contract. Fitting over a MICROCOSM frame reads the frame's typed survey weights by default; fitting over a plain data frame — the standalone path — requires the weights explicitly (a weight column name, a weight vector, or the literal `"none"`), and omitting them is an error rather than a silent unweighted fit. The contract makes the weight-blindness failure mode of Section 2.3 unrepresentable by accident.

3.2 Ablations and baselines

Table 1 lays out the method surface. Two design choices are evaluated by knocking them out with everything else held fixed: the full estimator versus itself with `weights="none"` (the weighted bootstrap's contribution), and versus fitting each target independently (chaining's contribution). Gating has no such knockout in this surface: `microcosm-fit` does not expose a gates-off switch, and the nearest ungated comparator, the `microimpute` forest, differs from the full estimator in more than the gate. It chains its targets as the full estimator does, but it turns each requested quantile level into a draw from a ten-level conditional quantile grid (levels $1/11$ to $10/11$) at a randomly chosen index. Under the level sampling used for all baselines here, that draws the lowest and highest levels 25 and 21 percent of the time, against

Method	Weights	Gates	Chaining	Implementation
`microcosm-fit`	design	yes	yes	`microcosm-fit`
– unweighted	none	yes	yes	`microcosm-fit`, `weights="none"`
– unchained	design	yes	no	`microcosm-fit`, targets fit independently
QRF (`microimpute`)	design	no	yes	`microimpute` QRF, as shipped
OLS	design	no	no	`microimpute`
Linear quantile regression	design	no	no	`microimpute`
NND hot deck	standardization only	via donation	via donation	`py-statmatch`
Weighted marginal draw	design	—	—	unconditional weighted draw

Table 1: The method surface. Every fit that has a weights column uses the survey's design weights; the one exception is the unweighted ablation, whose sole difference from the full estimator is `weights="none"`. The `microimpute` forest is an external baseline run as shipped. It keeps the design weights and chains its targets, but it has no gate and it draws through its own sampler, so it is not a knockout of the full estimator. Hot deck donates all target values from one matched donor record, which preserves zero masses and cross-target structure by construction rather than by modeling them; its donor weights standardize the matching variables and do not enter the choice of donor.

10 percent each for a uniform draw. Its random state is also fixed. Contrasts with it are therefore comparisons against standard practice, not attributions to gating. Baselines are standard practice: quantile-regression-forest, ordinary-least-squares, and linear quantile-regression imputers from `microimpute` (PolicyEngine, 2025), and nearest-neighbour-distance hot-deck statistical matching from `py-statmatch`, a Python port of the R package StatMatch (D'Orazio, 2025). The full estimator and the `microimpute` forest are built on the same forest library (Johnson, 2024).

3.3 The population-view harness

Per-variable metrics cannot adjudicate the question this paper cares about, because the methods differ most in how they treat the *joint*. We therefore evaluate within a framework we call the population-view harness, sketched in Figure 1.

The framework has three ingredients. One latent population distribution P produces individuals. Each survey s observes it through a *view* V_s — a variable subset, a sampling design, a measurement idiom — and the survey's holdout H_s is a weighted sample from $V_s(P)$ (Lohr and Raghunathan, 2017). A candidate weighted file Q (any generator's) is scored by projecting it through each view and comparing $V_s(Q)$ to H_s *in the view's own variable space*. Cross-survey consistency is never required — the candidate is only asked to explain each view in its own idiom — which keeps the harness usable even though surveys disagree

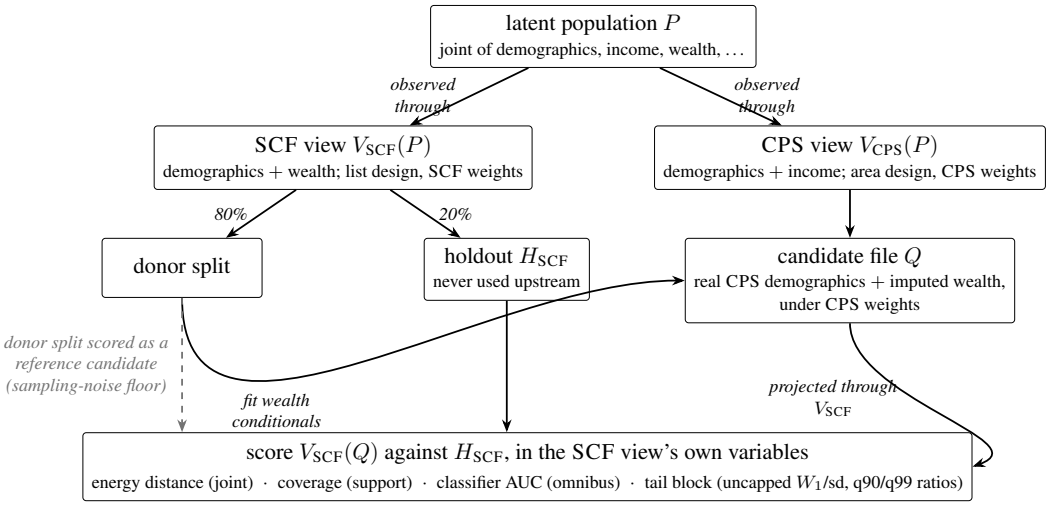


Figure 1: The population-view harness. One latent population is observed through survey views — here the SCF (demographics and wealth, list design) and the CPS (demographics and income, area design). A candidate file assembles real CPS demographics with imputed wealth; the harness projects it through the SCF view and scores it against a held-out SCF sample on four blocks. The holdout appears nowhere upstream of any candidate, and the donor split itself is scored as a reference candidate to calibrate sampling noise.

with one another about the population they share.

Each view carries four complementary blocks, chosen so that no single failure mode can hide. For weighted samples $X = \{(x_i, a_i)\}$ from the candidate and $Y = \{(y_j, b_j)\}$ from the holdout (weights normalized to sum to one):

Energy distance (joint fit; lower is better, 0 is perfect). The weighted energy distance (Székely and Rizzo, 2013)

$$\mathcal{E}(X,Y) = 2\sum_{i,j} a_i b_j \|x_i - y_j\| - \sum_{i,i'} a_i a_{i'} \|x_i - x_{i'}\| - \sum_{j,j'} b_j b_{j'} \|y_j - y_{j'}\|$$

is the sample form of a strictly proper scoring rule (Gneiting and Raftery, 2007): its expectation is uniquely minimized by the true distribution, so a candidate hedged toward modal households scores strictly worse than one matching the full joint.

Coverage (support; higher is better, 1 is perfect). PRDC coverage (Naeem et al., 2020) is the weighted fraction of holdout points with at least one candidate point inside their k-nearest-neighbour radius — does the candidate reach every neighbourhood the holdout

occupies? It is the explicit anti-collapse axis, and because it depends on the candidate only through its *support*, it is by definition invariant to any reweighting of the candidate: the one block a later calibration cannot change. As implemented, each sample larger than 2,048 points is first reduced to that size by a weight-proportional resample (the cap the energy distance shares). The candidate always is, so the reported coverage is invariant to reweighting only up to that resample.

Classifier two-sample test (omnibus; 0.5 is indistinguishable). A gradient-boosted classifier is trained to tell candidate rows from holdout rows at equal class mass, and we report its cross-validated AUC (Lopez-Paz and Oquab, 2017). This is the omnibus block — a single overall test sensitive to *any* difference in the joint distribution, including dependence violations that leave every marginal intact. An AUC of 0.5 means the classifier cannot separate the files; 1.0 means it always can.

Tail block (extremes; ratio of 1 is perfect). The fourth block exists because we caught the first three missing a real failure. Pairwise sample-geometry metrics run on capped subsamples, and a candidate whose imputed wealth 99th percentile is double the holdout's can still tie on energy distance: the discrepant mass is a sliver of standardized pairwise space. Economic variables live in heavy right tails, so each *imputed* column also carries a tail block computed on the full weighted samples with no cap: its weighted Wasserstein-1 distance to the holdout, scaled by the holdout's weighted standard deviation, and its weighted 90th- and 99th-percentile ratios (candidate over holdout).

Holdouts are used nowhere upstream, so the harness is a non-self-referential test surface. A sampling-noise floor — the donor split itself scored as a candidate — anchors what "as good as another sample of the survey" means on each axis, including how much a tail ratio wobbles under sampling noise alone.

3.4 Marginal and stress metrics

All metrics are computed on held-out receiver rows under survey weights, with one exception noted under pinball loss.

Pinball loss. Pinball loss — also called quantile loss; the terms are interchangeable and we use *pinball* throughout — scores predicted quantiles against realized values (Koenker, 2005). For a predicted quantile $\hat{q}_\tau$ and realized value y, the loss is $\rho_\tau(y - \hat{q}_\tau)$ with $\rho_\tau(u) =$

$u\,(\tau - \mathbf{1}[u < 0])$; we average it, weighted, over the decile grid $\tau \in \{0.1, \ldots, 0.9\}$. A single stochastic draw carries no conditional quantiles, so $\hat{q}_\tau$ here is the τ-quantile of the method's pooled receiver draw, read with every receiver row counted once, and it is scored against every held-out value under the receiver weights. As computed, pinball loss is therefore a *marginal* score. It does not test conditional calibration and cannot separate a conditional method from an unconditional one. Because the predicted quantiles are read without weights, it also favors a draw whose unweighted quantiles already match the weighted population.

Wasserstein-1 distance. Marginal fit is scored by the weighted Wasserstein-1 distance to the weighted donor distribution,

$$W_1(F, G) = \int_0^1 \left|F^{-1}(u) - G^{-1}(u)\right| du,$$

the average absolute gap between the two distributions' quantile functions — equivalently, the area between their CDFs. The subscript is the order: W_1 averages absolute quantile gaps in the variable's own units (dollars here), where W_2 would square them. We use order 1 because it is interpretable in dollars and does not let a single extreme quantile dominate the score quadratically.

Zero-share error. The absolute difference between the candidate's and the donor's weighted share of exact zeros — preservation of the zero mass of zero-inflated targets.

Reweighting fragility. Fragility formalizes the landmine diagnostic of Section 1: over the family of bounded multiplicative reweightings $w_i \mapsto m_i w_i$ with $m_i \in [1/\kappa, \kappa]$ (the hard weight-ratio bounds production calibration guards use), the worst-case single-record share of a target aggregate has the closed form $\kappa^2 c^* / (\kappa^2 c^* + S - c^*)$, where c^* is the largest baseline contribution $w_i|a_i|$ and S their sum. Read it as: after the most adversarial reweighting a calibrator could apply, what share of the aggregate can one record carry? We report it at $\kappa = 5$ for every method next to the held-out truth's own fragility — the exposure a faithful method should *match*, not minimize.

3.5 Protocol

Every task uses an 80/20 donor/receiver holdout repeated over ten seeds, with splits paired across methods: the split is a pure function of the seed, so every method fits and is scored on identical partitions. Task inputs are frozen once and pinned by hash; only the method

varies within a sweep. All results tables are generated from the sweep artifacts by the repository's command line; no result is hand-entered. Where a method is unavailable or a metric undefined for a task, the cell is reported as absent rather than omitted.

4 Data and tasks

Four task families span the settings the estimator is used in. Where a task imputes within one survey, the protocol of Section 3.5 applies: 80/20 donor/receiver holdouts repeated over ten paired seeds. The population-view experiment additionally projects candidates onto a second survey. Every table in Section 5 regenerates from the repository's committed run configurations; row caps and skipped cells are recorded in the run manifests.

4.1 Task 1: wealth within the SCF

The Survey of Consumer Finances (Board of Governors of the Federal Reserve System, 2023) observes household wealth in detail. From the 2022 summary extract (first implicate, survey weights) we impute `debt` (zero-inflated and nonnegative) and `networth` (sign-mixed, with a heavy right tail) from age, sex, education class, marital status, children, total income, and wage income. The two targets are chosen deliberately: their sign structures exercise the regime gates, and their strong mutual dependence exercises chaining.

4.2 Task 2: the SCF→CPS population view

Each method fits wealth conditionals on an SCF donor split and imputes them onto real CPS ASEC households sharing the predictor set — households built from the Census Bureau's ASEC public-use files (U.S. Census Bureau, 2025), read directly from `census.gov` so that no task input flows through any PolicyEngine processing. The experiment runs in two profiles. The *minimal* profile — six shared predictors, the two wealth targets of Section 4.1, a single receiver vintage (ASEC 2025) — is the controlled instantiation in which every mechanism is legible. The *production-scale* profile mirrors the production stack's shape in three ways. The shared predictor set widens to everything the two surveys genuinely share: ten variables, adding education class, race, homeownership, and labor-force status, each mapping verified against both files' codebooks. The receiver pools three ASEC vintages (2023–2025, weights scaled to one household population, 168,852 households), as the production support spine does. And the targets widen to a four-component chained balance sheet — financial assets, nonfinancial assets, debt, then net worth — whose accounting

identity networth = fin + nfin − debt holds exactly in the extract, so chaining's contribution is directly interpretable. In both profiles each run imputes onto a fixed uniform subsample of 20,000 receiver households (of 55,762 in the minimal profile and 168,852 at production scale), the same for every method and seed. The candidate population — CPS demographics plus imputed wealth, under CPS household weights — is then scored through the harness of Section 3.3 against the *held-out* SCF: the joint of shared predictors and wealth, under SCF weights. The `scf_sample_reference` rows score the donor split itself against the holdout, anchoring the sampling-noise floor. This is the paper's direct test of whether conditional-distribution methods preserve joint structure that matching's conditional-independence gluing loses (Rässler, 2002; D'Orazio et al., 2006), measured where it matters: on a receiver file the donor survey never saw. Because the demographic block of every candidate is the same real CPS sample, differences between methods isolate the wealth block and its coupling to demographics; the shared demographic offset between the two surveys is common to all methods and bounded below by the reference floor.

4.3 Task 3: zero-inflated income components in the CPS

From the Census Bureau's ASEC 2025 public-use person file (income reference year 2024), read directly from `census.gov` — never a processed or enhanced artifact, which would embed the production imputations this paper is about — we impute interest and dividend income for adults from age, sex, and employment income, under the ASEC person supplement weights. (Survey files publish per-person design weights; MICROCOSM's production convention deliberately carries calibrated *household* weights onto persons, since its calibration operates at household grain. The task fixes the survey's own published weights, and because every method shares whatever weights the task fixes, the method comparison does not turn on this choice.) Interest is zero for 41.7 percent of adults and dividends for 84.7 percent (weighted shares), both with heavy positive tails — the regime surface the gates are built for. The task runs on a fixed uniform subsample of 20,000 of the file's 108,044 adults. Public-use ASEC amounts are topcoded, which truncates the extreme tail relative to an administrative or SCF-style source; the fragility comparisons of Section 5.2 are therefore conservative. This task also carries the reweighting stress test of Section 3.4. The test is motivated by prior POLICYENGINE production experience, in which an unweighted fit broadcast rare high-value donor records across a receiver file as near-zero-weight point masses that later reweighting amplified. All fragility numbers reported here are measured under this paper's protocol.

Method	Debt pinball	Debt W_1	Debt zero-share err.	Net worth pinball	Net worth W_1
microcosm-fit	49,201 (2,637)	**13,001** (6,285)	0.020 (0.012)	551,317 (36,012)	**135,871** (51,106)
– unweighted	50,218 (2,452)	32,247 (16,930)	**0.012** (0.014)	703,457 (77,173)	829,158 (445,108)
– unchained	49,201 (2,637)	**13,001** (6,285)	0.020 (0.012)	556,622 (23,349)	158,606 (44,310)
QRF (microimpute)	50,433 (3,120)	16,619 (5,358)	0.018 (0.018)	560,819 (36,903)	234,367 (142,057)
OLS	70,053 (17,637)	156,295 (74,822)	0.226 (0.018)	1,116,572 (383,007)	3,565,923 (2,062,063)
Quantile regression	49,797 (3,391)	29,460 (20,456)	0.151 (0.031)	534,420 (37,801)	295,470 (110,986)
NND hot deck (py-statmatch)	49,537 (2,627)	13,770 (5,141)	0.017 (0.013)	644,921 (51,255)	509,013 (399,757)
Weighted marginal draw	**48,277** (2,586)	14,617 (4,732)	0.020 (0.017)	**443,554** (20,651)	215,398 (87,264)

Table 2: Wealth imputation within the SCF (debt and net worth from demographics and income): weighted pinball loss, Wasserstein-1 to the weighted donor distribution, and zero-share error, mean (sd) over ten paired seeds. Bold marks the best non-reference value per column. Lower is better.

4.4 Task 4: cross-dataset transfer

To test behavior beyond economic microdata, we run the method surface over the six OpenML AutoML Benchmark regression datasets used in the `microimpute` manuscript's appendix (Vanschoren et al., 2014) — `space_ga`, `elevators`, `brazilian_houses`, `onlinenewspopularity`, `abalone`, and `house_sales` — treating each dataset's target as the imputed column under uniform weights. The two datasets with more than 20,000 rows, `house_sales` and `onlinenewspopularity`, are capped at 20,000 by a fixed uniform subsample.

5 Results

Throughout, *the full estimator* names `microcosm-fit` with all three design choices on (weighted, gated, chained); each table indents its two ablations beneath it. The `microimpute` forest, OLS, quantile regression, and hot deck are the standard-practice baselines (Table 1), and the weighted marginal draw is the unconditional lower bound. Unless a caption says otherwise, every cell is the mean over the ten paired seeds with the standard deviation across seeds in parentheses, and bold marks the best non-reference value in a column: lowest for distances and losses, highest for coverage, closest to 0.5 for the two-sample AUC, closest to 1 for tail ratios, and closest to the held-out truth's own value for fragility.

5.1 Wealth within the SCF

Table 2 reports the within-SCF task. The headline is the weighting ablation: switching the full estimator to `weights="none"` degrades net-worth Wasserstein-1 by a factor of six (135,871 to 829,158) and debt by a factor of 2.5 (13,001 to 32,247). The SCF's list-sample

design oversamples wealthy households at low weight, so an unweighted fit learns the sample's wealth distribution rather than the population's — this is the weight-blindness failure mode of Section 1 measured directly, and no other design choice comes close to its effect size. The other contrasts are smaller: the `microimpute` forest sits at 234,367 on net-worth W_1 (1.7 times the full estimator), and removing chaining raises net-worth W_1 by 17 percent (135,871 to 158,606).

Against the baselines, the full estimator's margins are largest where tails and joints matter. Hot-deck matching nearly ties it on debt marginal fit (13,770 vs. 13,001) but is 3.7 times worse on net worth (509,013). Its donor selection does not use the weights (Table 1), and its loss runs in the same direction as the unweighted ablation's, so this contrast does not separate matching from weight-blindness. OLS is catastrophic on both (156,295 and 3.6 million): normal residuals cannot represent a wealth distribution. Linear quantile regression leaves 15 percent of the debt zero mass unaccounted for, against at most 2 percent for the forest-based methods.

One result reads wrong until the metric is examined: the *unconditional* weighted marginal draw posts the best pinball loss on both targets. Pinball loss as computed here is a marginal score whose predicted quantiles are read without receiver weights (Section 3.4). In a survey that oversamples the wealthy, a conditional method's pooled draw follows the receiver sample's composition, while the marginal draw is centered on the weighted population by construction, so this column favors the marginal draw for a reason unrelated to conditional fit, and we do not read its lead as evidence about the methods. More generally, no receiver-side marginal metric can distinguish a method that draws from the correct weighted marginal while ignoring predictors from one that also gets the conditionals right. That is what the population-view harness of Section 5.5, which scores the joint, is for.

5.2 Zero-inflated components and reweighting fragility

Table 3 reports the CPS components task. Pinball loss barely separates the non-OLS methods — as a marginal score it is not built to — but the zero mass separates them sharply. The full estimator holds zero-share error to 0.6 and 1.3 percentage points (dividends, interest); the `microimpute` forest, which has no gate, drifts to 4.4 points on dividends, a contrast that does not isolate gating (Section 3.2); linear quantile regression misses by 25–27 points because a 99-level linear quantile grid cannot hold a five-sixths zero mass; and OLS misses the dividend zero share by 84 points — with normal residual draws, nearly every adult becomes a dividend recipient. Hot deck, which donates observed values, holds the zero mass

Method	Interest pinball	Interest zero-share	Dividend pinball	Dividend zero-share
microcosm-fit	1,491 (63.58)	0.013 (0.011)	320.68 (45.26)	0.006 (0.004)
– unweighted	**1,491** (63.62)	0.011 (0.007)	320.68 (45.29)	0.008 (0.005)
– unchained	1,491 (63.58)	0.013 (0.011)	320.69 (45.26)	**0.006** (0.005)
QRF (microimpute)	1,493 (63.34)	0.014 (0.011)	322.17 (45.24)	0.044 (0.010)
OLS	2,527 (63.57)	0.422 (0.012)	961.52 (112.29)	0.844 (0.005)
Quantile regression	1,493 (63.23)	0.270 (0.016)	320.91 (45.19)	0.251 (0.011)
NND hot deck (py-statmatch)	1,493 (63.89)	0.021 (0.019)	320.94 (45.22)	0.009 (0.009)
Weighted marginal draw	1,491 (63.61)	**0.011** (0.013)	**320.68** (45.25)	0.006 (0.004)

Table 3: Zero-inflated income components in the raw CPS (interest and dividend income for adults, on a fixed subsample of 20,000 of the file's 108,044 adults): weighted pinball loss and zero-share error, mean (sd) over ten paired seeds. Bold marks the best non-reference value per column; lower is better.

Method	Interest fragility	(truth)	Dividend fragility	(truth)
microcosm-fit	0.384 (0.051)	0.372 (0.043)	0.675 (0.100)	0.614 (0.104)
– unweighted	0.395 (0.043)	0.372 (0.043)	0.699 (0.057)	0.614 (0.104)
– unchained	0.384 (0.051)	0.372 (0.043)	0.661 (0.100)	0.614 (0.104)
QRF (microimpute)	0.364 (0.064)	0.372 (0.043)	**0.633** (0.087)	0.614 (0.104)
OLS	0.055 (0.010)	0.372 (0.043)	0.055 (0.009)	0.614 (0.104)
Quantile regression	**0.368** (0.105)	0.372 (0.043)	0.444 (0.073)	0.614 (0.104)
NND hot deck (py-statmatch)	0.391 (0.027)	0.372 (0.043)	0.647 (0.140)	0.614 (0.104)
Weighted marginal draw	0.356 (0.042)	0.372 (0.043)	0.746 (0.067)	0.614 (0.104)

Table 4: Reweighting fragility at $\kappa = 5$: worst-case single-record share of each component's aggregate over bounded multiplicative reweightings, next to the held-out truth's own fragility. The target is the truth's exposure — bold marks the closest method; substantially lower means the method fails to generate realistic tails, substantially higher means it plants landmines.

within 0.9–2.1 points, as expected.

Table 4 reports the landmine diagnostic. Two readings matter. First, the held-out truth is itself concentrated: under $\kappa = 5$ reweighting a single record can carry 37 percent of the interest aggregate and 61 percent of the dividend aggregate at this sample size — heavy-tailed capital income is genuinely fragile, and a faithful method should *match* that exposure, not minimize it. Second, every forest- and donation-based method tracks the truth's fragility closely, while OLS sits far below it, at 0.06 against 0.37 and 0.61 — understating the true exposure by factors of roughly 7 and 11. Its aggregate is spread across many moderate records because the method never generates realistic tails; the zero-share and Wasserstein numbers show the same failure from other angles. No method exceeds the truth's exposure materially on this within-survey task; the landmine pathology of Section 1 arises when unweighted cross-file fits meet downstream reweighting, and the direct evidence for it here is the tail block of the population-view experiment (Section 5.5).

Task	Target	Metric	Δ unweighted	Δ unchained	Δ QRF (microimpute)
cps components	dividend_income	pinball_loss	0.000 (0.056)	0.005 (0.033)	1.48 (0.327)
cps components	dividend_income	wasserstein1	36.11 (46.53)	11.29 (94.58)	308.65 (169.93)
cps components	dividend_income	zero_share_error	0.002 (0.006)	-0.000 (0.006)	0.038 (0.012)
cps components	interest_income	pinball_loss	-0.016 (0.307)	0.000 (0.000)	1.57 (1.35)
cps components	interest_income	wasserstein1	-26.28 (114.17)	0.000 (0.000)	116.69 (203.92)
cps components	interest_income	zero_share_error	-0.002 (0.006)	0.000 (0.000)	0.001 (0.007)
scf wealth	debt	pinball_loss	1,017 (709.88)	0.000 (0.000)	1,232 (713.52)
scf wealth	debt	wasserstein1	19,246 (16,086)	0.000 (0.000)	3,618 (5,388)
scf wealth	debt	zero_share_error	-0.008 (0.011)	0.000 (0.000)	-0.001 (0.021)
scf wealth	networth	pinball_loss	152,140 (51,569)	5,305 (23,838)	9,502 (29,097)
scf wealth	networth	wasserstein1	693,287 (431,527)	22,735 (70,684)	98,496 (179,722)
scf wealth	networth	zero_share_error	-0.000 (0.001)	0.000 (0.001)	0.001 (0.003)

Table 5: Paired-seed deltas from the full estimator, mean (sd) over ten seeds, for its two ablations and for the `microimpute` forest baseline. Positive means the comparison row is worse on these lower-is-better metrics. For the two ablations, a delta near zero means the knocked-out design choice is inert on that task; the forest baseline differs from the full estimator in more than one respect, so its column is a comparison, not an attribution.

5.3 Ablations: attributing the gains

Table 5 attributes the full estimator's behavior to the two design choices it can knock out, and sets the `microimpute` forest beside them. Each ablation cell answers one question: on the same ten splits, how much did this metric worsen when one design choice was knocked out? The $+693{,}287$ under Δ unweighted on net-worth W_1, for instance, reads: refitting the identical estimator without weights raised the marginal-fit distance by \$693,287 on average. Because ablation and estimator share each split, the delta subtracts out split-to-split noise; the parenthetical standard deviation says how much the delta itself varies across the ten splits.

The ordering is task-dependent in an informative way. On the SCF — informative weights, heavy tails — weighting dominates ($+693{,}287$ net-worth W_1 when removed), and chaining contributes a smaller $+22{,}735$ mean shift, though with wide seed-to-seed dispersion (sd $70{,}684$); the `microimpute` forest trails by $+98{,}496$ (sd $179{,}722$). On the CPS components — weakly informative weights, extreme zero inflation — weighting has a small effect (removing it raises dividend W_1 by 36, from 102, on eight of the ten seeds, and leaves the other metrics within seed noise) and chaining is inert. The one large contrast there is with the `microimpute` forest ($+309$ dividend W_1, and the zero-share drift of Table 3), which this surface cannot attribute to gating alone. Which choice binds depends on the survey's design and the target's structure. Where a choice does not bind, its cost is small: the unweighted ablation is slightly ahead on SCF debt zero share (0.012 against 0.020, Table 2) and on the uniform-weight OpenML suite (Section 5.4).

Method	abalone	brazilian_houses	elevators	house_sales	onlinenewspopularity	space_ga	Mean rank
microcosm-fit	0.254	166.64	0.000	7,080	274.67	0.008	3.50
– unweighted	0.257	184.92	0.000	6,744	233.10	0.008	**3.00**
– unchained	0.254	166.64	0.000	7,080	274.67	0.008	3.50
QRF (microimpute)	0.174	191.36	0.000	8,461	546.16	0.012	4.50
OLS	0.532	212.09	0.002	90,518	6,117	0.015	7.83
Quantile regression	0.321	212.04	0.001	24,115	382.98	0.085	7.00
NND hot deck (py-statmatch)	0.176	199.12	0.000	10,212	161.43	0.010	3.50
Weighted marginal draw	0.148	210.50	0.000	7,135	143.52	0.011	3.17

Table 6: Wasserstein-1 to the donor distribution across the six OpenML regression datasets (uniform weights), mean over ten paired seeds, with mean rank across datasets among the eight methods shown. The two datasets with more than 20,000 rows (`house_sales`, 21,613, and `onlinenewspopularity`, 39,644) are capped at 20,000 by a fixed uniform subsample.

5.4 Cross-dataset transfer

Table 6 runs the surface over the six OpenML regression datasets under uniform weights. Two observations. First, with weights uninformative by construction, the full estimator and its unweighted ablation differ only by a uniform resample of the training rows, and they rank near the top (mean ranks 3.5 and 3.0) alongside the unconditional marginal draw (3.2); the linear methods rank last (7.0 and 7.8). Second — and more important for this paper's argument — a method with no conditional structure at all ranks second of the eight on a *marginal* distance, ahead of the full estimator and of every baseline. Marginal metrics saturate: they reward drawing from the right marginal and cannot rank what imputation is actually for. The population-view harness exists because of exactly this ceiling.

5.5 The SCF→CPS population view

Tables 7 and 8 report the population-view experiment in its two profiles. The floor behaves as designed in both: another sample of the same survey is statistically indistinguishable (energy 0.009 and 0.012, coverage 0.987 and 0.986, AUC 0.496 and 0.499), and its q99 ratio of 1.10 calibrates how much a 99th-percentile ratio moves under sampling noise alone. Four findings follow.

First, *the weighting failure is a tail failure, and only the tail block sees it*. The unweighted ablation ties the full estimator on energy distance, coverage, and the C2ST in both profiles — and its imputed net-worth q99 is 1.93 times the holdout's in the minimal profile and 2.25 times at production scale, with tail W_1/sd five times the full estimator's (0.159 vs. 0.031 minimal; 0.133 vs. 0.027 at scale). Fitting the SCF unweighted learns the wealthy oversample's conditionals, and the resulting inflation concentrates beyond the resolution of capped pairwise geometry: we report this as a finding about evaluation practice. Any

Method	Energy distance	Coverage	C2ST AUC	NW W_1/sd	NW q99 ratio
SCF sample (floor)	0.009 (0.006)	0.987 (0.009)	0.496 (0.023)	0.013 (0.003)	1.10 (0.202)
microcosm-fit	**0.139** (0.026)	0.960 (0.014)	0.732 (0.013)	0.031 (0.007)	0.696 (0.112)
– unweighted	0.140 (0.026)	0.958 (0.014)	**0.731** (0.016)	0.159 (0.041)	1.93 (0.255)
– unchained	0.139 (0.026)	0.958 (0.016)	0.742 (0.013)	0.029 (0.009)	0.765 (0.177)
QRF (microimpute)	0.140 (0.027)	0.957 (0.012)	0.775 (0.008)	0.029 (0.007)	0.805 (0.156)
OLS	0.166 (0.045)	0.732 (0.087)	0.988 (0.005)	0.434 (0.282)	**0.967** (0.505)
Quantile regression	0.141 (0.027)	0.938 (0.028)	0.927 (0.018)	0.052 (0.013)	0.530 (0.118)
NND hot deck (py-statmatch)	0.141 (0.027)	**0.962** (0.013)	0.753 (0.014)	0.051 (0.016)	0.762 (0.148)
Weighted marginal draw	0.139 (0.026)	0.937 (0.005)	0.848 (0.008)	**0.014** (0.004)	1.09 (0.222)

Table 7: The population-view experiment, *minimal* profile (six shared predictors, two targets, a fixed subsample of 20,000 of the 55,762 ASEC 2025 receiver households): candidates assembled from real CPS demographics plus each method's imputed wealth, under CPS household weights, scored against the held-out SCF through the SCF view. The first row is the sampling-noise floor; cells are mean (sd) over the ten paired donor splits. Energy distance lower is better; coverage higher; C2ST AUC of 0.5 is indistinguishable; net-worth W_1/sd lower is better; a q99 ratio of 1 is a perfect extreme tail. Bold marks the best non-reference value per column under those conventions.

Method	Energy distance	Coverage	C2ST AUC	NW W_1/sd	NW q99 ratio
SCF sample (floor)	0.012 (0.005)	0.986 (0.003)	0.499 (0.021)	0.013 (0.003)	1.10 (0.202)
microcosm-fit	**0.118** (0.024)	0.969 (0.009)	0.761 (0.026)	0.027 (0.008)	0.767 (0.190)
– unweighted	0.120 (0.024)	**0.971** (0.008)	**0.754** (0.011)	0.133 (0.038)	2.25 (0.443)
– unchained	0.118 (0.024)	0.967 (0.011)	0.819 (0.015)	0.028 (0.006)	0.699 (0.112)
QRF (microimpute)	0.126 (0.025)	0.965 (0.011)	0.817 (0.009)	0.033 (0.022)	1.15 (0.445)
OLS	0.152 (0.059)	0.877 (0.055)	0.995 (0.002)	0.435 (0.282)	**0.963** (0.503)
Quantile regression	0.120 (0.026)	0.951 (0.011)	0.970 (0.007)	0.051 (0.013)	0.485 (0.098)
NND hot deck (py-statmatch)	0.119 (0.025)	0.971 (0.010)	0.768 (0.009)	0.024 (0.008)	0.747 (0.107)
Weighted marginal draw	0.119 (0.024)	0.931 (0.015)	0.968 (0.003)	**0.015** (0.003)	1.12 (0.241)

Table 8: The population-view experiment, *production-scale* profile (ten shared predictors, four chain-ordered balance-sheet targets, a fixed subsample of 20,000 of the 168,852 pooled ASEC 2023–2025 receiver households). Columns, cell format, and bolding as in Table 7.

harness whose joint metrics subsample — which at these sample sizes they must — needs an uncapped tail block for heavy-tailed economic variables, or it will certify files whose top percentile is wrong by a factor of two.

Second, *chaining's contribution appears only where the targets are jointly constrained — and then only a joint metric sees it*. In the minimal profile (two targets) the unchained ablation is indistinguishable from the full estimator. At production scale, where the four targets satisfy the accounting identity networth = fin + nfin − debt exactly in the donor, fitting them independently costs 5.8 points of C2ST AUC (0.819 vs. 0.761) while leaving every marginal and tail statistic essentially unchanged — the classifier detects balance sheets that do not add up, which no marginal metric can. The `microimpute` forest sits at the same

level (0.817). It differs from the full estimator in more than one respect (Section 3.2), so this experiment credits no part of that gap to gating.

Third, *richer conditioning makes the classifier two-sample test — the scorecard's omnibus block — decisive*. The unconditional weighted marginal draw — at or near the top on marginal metrics throughout this paper — moves from AUC 0.848 in the minimal profile to 0.968 at production scale, while the full estimator stays the closest to the floor after its own unweighted ablation (0.732 and 0.761, against 0.731 and 0.754). With ten shared predictors and a four-component wealth block, a classifier finds the missing demographics–wealth coupling almost perfectly. This is the empirical case for carrying strong auxiliary predictors in a production imputation stack: richer conditioning is what turns joint-structure failures from undetectable into obvious.

Fourth, *hot-deck matching is a genuine near-peer on this task*. It sits within noise of the full estimator on energy and coverage in both profiles, within 2.1 and 0.7 AUC points, and its donated values slightly beat the full estimator's on tail W_1 at scale (0.024 vs. 0.027). The full estimator's advantages over matching live where this task does not press hard: informative-weight regimes (Table 2, where the matching baseline also selects donors without weights), extreme zero inflation (Table 3), and multi-target coherence (the chaining result above, which matching handles only through whole-record donation at the cost of reusing donors). One more tail observation cuts against the full estimator: its imputed net-worth q99 falls short of the holdout's, at a ratio of 0.70 in the minimal profile and 0.77 at production scale. Hot deck, which donates observed values, shows the same shortfall (0.76 and 0.75), while the unconditional marginal draw, which draws from the donor's weighted marginal without conditioning, sits near 1 (1.09 and 1.12, against a floor of 1.10). The shortfall is therefore shared by a forest and a donation method that both condition on the receiver's demographics, and this experiment does not isolate its cause. No method in this surface gets the extreme tail right for free; the tail block is what keeps that visible.

6 Discussion

The results attribute the estimator's behavior to its design choices with a regularity worth stating plainly. *Weighting is the dominant choice wherever the design is informative*: on the SCF, whose list sample oversamples wealthy households, removing the weighted bootstrap costs a factor of six on net-worth marginal fit within the donor survey and inflates the imputed 99th percentile on the population view by a factor of 1.9 to 2.3. That effect is larger than

the gap between the full estimator and every competing method except OLS. On the CPS components task, whose person weights carry little outcome information, weighting matters far less: removing it raises dividend W_1 by about a third and leaves the other metrics within seed noise. An estimator cannot know in advance which regime it is in, which is the argument for weighting by construction rather than by option. *Gating is the choice this surface tests least directly*: the full estimator holds dividend zero-share error to 0.6 percentage points, against 4.4 for the ungated `microimpute` forest, but that baseline also draws through a different sampler, and no gates-off arm of the full estimator is run. *Chaining matters only where the targets are jointly constrained*: invisible with two loosely coupled targets, it is worth 5.8 AUC points when the four imputed components must satisfy a balance-sheet identity — and only the omnibus classifier sees the violation, because a balance sheet that does not add up has perfectly reasonable marginals. Where a choice does not bind, its cost is small: the unweighted ablation is slightly ahead on SCF debt zero share and on the uniform-weight OpenML suite.

The evaluation findings are as load-bearing as the method findings, and one of them corrected this paper's own harness. Marginal metrics saturate: with weak predictors, an unconditional weighted draw from the donor marginal sits at or near the top on pinball loss and Wasserstein distance — on the OpenML suite it ranks second of the eight methods — while carrying no conditional structure at all; at production scale the classifier separates it almost perfectly (AUC 0.968 against 0.761). But the joint metrics have their own blind spot: capped pairwise geometry tied the unweighted ablation with the full estimator while its imputed net-worth 99th percentile was double the holdout's, because the discrepant mass is a sliver of standardized pairwise space. We caught this by comparing weighted quantiles directly, added the uncapped tail block to the harness, and report it as a general lesson: sample-geometry evaluations of heavy-tailed economic microdata need an explicit tail axis, with a sampling floor to calibrate its noise (the floor's own q99 ratio wobbles to 1.10). The fragility diagnostic adds the complementary caution: the target is the truth's own tail exposure, not minimal exposure, and the one method that minimizes it (OLS, at factors of 7 to 11 below truth) does so by failing to generate realistic tails at all.

Several limitations bound the claims. First, the population-view experiment runs one view pair: the common CPS–SCF offset (differing income concepts, ASEC topcoding, and income reference years spanning 2021 to 2024 across the pooled vintages) is shared by all methods but bounds how close any candidate can come to the floor. A multi-view instantiation — one candidate scored simultaneously against CPS, SIPP (U.S. Census

Bureau, 2023), and PSID (Institute for Social Research, University of Michigan, 2023) views — would tighten the identification and is the production release-gate use of the harness; we leave it to future work (PSID access requires registration), since the two-view experiment already identifies the joint-structure differences this paper is about. Second, the joint-geometry blocks still run on weight-proportional subsamples (2,048 points per side); the tail block and the C2ST, which use all rows, are the axes least affected. Third, the full estimator understates the extreme tail on the population view: its imputed net-worth q99 is 0.70 to 0.77 of the holdout's. Hot-deck matching shows the same shortfall (0.75 to 0.76) and the unconditional marginal draw does not (1.09 to 1.12), so the cause is not specific to forest draws; this experiment does not isolate it. The tail block keeps the shortfall visible, and it is a caution against reading the full estimator's wins as uniform. The OLS and quantile-regression baselines are additionally converted to samplers through a 99-level grid, truncating draws beyond the 1st and 99th conditional percentiles. Fourth, the `microimpute` forest is run as shipped. It draws each value from a ten-level conditional quantile grid, which reaches no further than the 9th and 91st conditional percentiles and over-draws those two levels, and its constructor takes no seed, so its forest and its sampler run on one fixed internal seed in every cell. Its paired-seed dispersion therefore carries no forest noise, and its contrasts with the full estimator mix the gate with the sampler. Fifth, comparisons hold shared forest backends and default hyperparameters fixed rather than tuning each method per task; the two ablations are exact controls, the cross-family comparisons are defaults-versus-defaults. Finally, we evaluate single imputations distributionally and do not propagate imputation uncertainty in the multiple-imputation sense (Rubin, 1987); for the population statistics microsimulation consumes, the distributional view is the relevant one, but inference from imputed files would require the fuller treatment.

7 Conclusion

Imputation sits upstream of every population statistic a microsimulation model produces, and the imputed files are reweighted by consumers the imputer never sees. We presented an estimator that treats survey weights as a structural obligation rather than an option — weighted bootstrap into the forests, directly weighted regime gates, sequential chaining, and an interface that refuses a silent unweighted fit — and a population-view harness that evaluates any candidate population against held-out survey views with a strictly proper joint score, a support-based coverage axis, a classifier two-sample test, and an uncapped tail block.

Empirically, the interface rule is the paper's largest single effect. On a survey whose design is informative, the identical estimator fit without weights is six times worse on net-worth marginal fit, and on the population view it inflates the imputed 99th percentile by a factor of 1.9 to 2.3. Capped joint geometry certifies that failure; only the tail block catches it, which is itself the paper's central lesson about evaluating imputed files. Chaining matters where targets are jointly constrained (5.8 AUC points on a balance-sheet identity at production scale). The gated estimator holds the zero mass of zero-inflated components to about one percentage point, where an ungated forest baseline misses the dividend zero share by 4 points and linear quantile regression misses both by 25 to 27; no gates-off arm isolates the gate's own share of that. And marginal metrics — the field's default — rank an unconditional marginal draw at or near the top throughout, while the harness separates it decisively. The estimator is available standalone as `microcosm-fit`; the harness, task configurations, and every number in this paper regenerate from the accompanying repository.

Conflict of interest

All authors are affiliated with POLICYENGINE, the nonprofit organization that develops MICROCOSM, `microcosm-fit`, and `microimpute`, the software evaluated in this paper. The authors have no other competing interests.

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Data and code availability

The paper and experiment code are at https://github.com/PolicyEngine/imputation-paper; every results table regenerates from the committed run artifacts via the repository's command line. The estimator is implemented in `microcosm-fit`, part of MICROCOSM, open source at https://github.com/PolicyEngine/microcosm. It is not on PyPI. It installs on its own, without the rest of the stack, as a Git dependency on the `packages/microcosm-fit` subdirectory of that repository; the paper repository's README gives the one-line command. Baseline implementations are `microimpute` (https://github.com/PolicyEngine/microimpute) and `py-statmatch` (https://github.com/CosilicoAI/py-statmatch). The reported sweeps ran against the version of `microcosm-fit` merged in commit `b538d8a` of the MICROCOSM

repository (then named `populace`; that commit ships the distribution as `populace-fit`, under `packages/populace-fit`), `microimpute` 1.15.1, and `py-statmatch` at commit `094a242`.

The inputs are public and need no registration: the SCF 2022 summary extract from the Federal Reserve (SHA-256 beginning `3bb4d890`); the Census Bureau's ASEC public-use bundles for 2023, 2024, and 2025 (the 2025 bundle, `asecpub25csv.zip`, has SHA-256 beginning `318845a2`); and the six OpenML datasets of Section 4.4, fetched by name at versions the loader asserts. The Federal Reserve replaced the 2022 extract at its published address on 9 October 2026 with a version re-based to 2025 dollars, so the file this paper used can no longer be downloaded from the agency. Each run directory's manifest records rows, caps, seeds, and methods.

References

119th United States Congress. SSI savings penalty elimination act, 2025. URL https://www.congress.gov/bill/119th-congress/senate-bill/1234. S. 1234 / H.R. 2540, introduced April 1, 2025.

Rebecca R. Andridge and Roderick J. A. Little. A review of hot deck imputation for survey non-response. *International Statistical Review*, 78(1):40–64, 2010.

Board of Governors of the Federal Reserve System. Changes in U.S. family finances from 2019 to 2022: Evidence from the Survey of Consumer Finances. Technical Report 4, Federal Reserve Bulletin, October 2023.

François Bourguignon and Amedeo Spadaro. Microsimulation as a tool for evaluating redistribution policies. *Journal of Economic Inequality*, 4(1):77–106, 2006.

Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

Jean-Claude Deville and Carl-Erik Särndal. Calibration estimators in survey sampling. *Journal of the American Statistical Association*, 87(418):376–382, 1992.

Marcello D'Orazio. *StatMatch: Statistical Matching or Data Fusion*, 2025. URL https://CRAN.R-project.org/package=StatMatch. R package version 1.4.3.

Marcello D'Orazio, Marco Di Zio, and Mauro Scanu. *Statistical Matching: Theory and Practice*. John Wiley & Sons, 2006.

Max Ghenis. Quantile regression: From linear models to trees to deep learning. Towards Data Science, 2018. URL https://medium.com/data-science/quantile-regression-from-linear-models-to-trees-to-deep-learning-af3738b527c3. Medium blog post.

Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007.

Jens Hainmueller. Entropy balancing for causal effects: a multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis*, 20(1):25–46, 2012.

Institute for Social Research, University of Michigan. Panel Study of Income Dynamics: Public use dataset. Ann Arbor, MI: Survey Research Center, 2023. URL https://psidonline.isr.umich.edu/.

Reid A. Johnson. quantile-forest: A Python package for quantile regression forests. *Journal of Open Source Software*, 9(93):5976, 2024. doi: 10.21105/joss.05976.

Arthur B. Kennickell. Multiple imputation in the survey of consumer finances. Technical report, Board of Governors of the Federal Reserve System, September 1998. Prepared for the August 1998 Joint Statistical Meetings, Dallas, TX.

Roger Koenker. *Quantile regression*. Cambridge University Press, 2005.

Roger Koenker and Gilbert J. Bassett. Regression quantiles. *Econometrica*, 46(1):33–50, 1978.

Diane Lambert. Zero-inflated Poisson regression, with an application to defects in manufacturing. *Technometrics*, 34(1):1–14, 1992.

Roderick J. A. Little and Donald B. Rubin. *Statistical analysis with missing data*. John Wiley & Sons, 2nd edition, 2002.

Sharon L. Lohr and Trivellore E. Raghunathan. Combining survey data with other data sources. *Statistical Science*, 32(2):293–312, 2017.

David Lopez-Paz and Maxime Oquab. Revisiting classifier two-sample tests. In *International Conference on Learning Representations*, 2017.

Nicolai Meinshausen. Quantile regression forests. *Journal of Machine Learning Research*, 7:983–999, 2006.

John Mullahy. Specification and testing of some modified count data models. *Journal of Econometrics*, 33(3):341–365, 1986.

Muhammad Ferjad Naeem, Seong Joon Oh, Youngjung Uh, Yunjey Choi, and Jaejun Yoo. Reliable fidelity and diversity metrics for generative models. In *International Conference on Machine Learning*, pages 7176–7185. PMLR, 2020.

PolicyEngine. Microimpute documentation, 2025. URL https://policyengine.github.io/microimpute/.

PolicyEngine. Microcosm (formerly Populace): A microsimulation dataset construction pipeline, 2026. URL https://github.com/PolicyEngine/microcosm. Software.

Trivellore E. Raghunathan, James M. Lepkowski, John Van Hoewyk, and Peter Solenberger. A multivariate technique for multiply imputing missing values using a sequence of regression models. *Survey Methodology*, 27(1):85–95, 2001.

Susanne Rässler. *Statistical Matching: A Frequentist Theory, Practical Applications, and Alternative Bayesian Approaches*. Springer, New York, 2002.

Donald B. Rubin. *Multiple imputation for nonresponse in surveys*. John Wiley & Sons, 1987.

Donald B Rubin. Statistical disclosure limitation. *Journal of Official Statistics*, 9(2):461–468, 1993.

Social Security Administration. SSI annual statistical report, 2024. Technical report, Social Security Administration, Office of Retirement and Disability Policy, 2024. URL https://www.ssa.gov/policy/docs/statcomps/ssi_asr/.

Holly Sutherland and Francesco Figari. EUROMOD: The European Union tax-benefit microsimulation model. *International Journal of Microsimulation*, 6(1):4–26, 2013.

Gábor J. Székely and Maria L. Rizzo. Energy statistics: A class of statistics based on distances. *Journal of Statistical Planning and Inference*, 143(8):1249–1272, 2013.

U.S. Census Bureau. Survey of Income and Program Participation (SIPP): 2022 panel. Washington, DC: U.S. Census Bureau, 2023. URL https://www.census.gov/programs-surveys/sipp.html.

U.S. Census Bureau. Current population survey, 2025 annual social and economic (ASEC)

supplement. Technical report, U.S. Census Bureau, 2025. URL https://www2.census.gov/programs-surveys/cps/datasets/2025/march/asecpub25csv.zip.

Stef Van Buuren and Karin Groothuis-Oudshoorn. mice: Multivariate imputation by chained equations in r. *Journal of Statistical Software*, 45(3):1–67, 2011.

Joaquin Vanschoren, Jan N. van Rijn, Bernd Bischl, and Luís Torgo. OpenML: Networked science in machine learning. *ACM SIGKDD Explorations Newsletter*, 15(2):49–60, 2014.

Paul T. von Hippel. Should a normal imputation model be modified to impute skewed variables? *Sociological Methods & Research*, 42(1):105–138, 2013. doi: 10.1177/0049124112464866.

Nikhil Woodruff and Max Ghenis. Enhancing survey microdata with administrative records: A novel approach to microsimulation dataset construction. Technical report, PolicyEngine, 2024.